



Department of Computer Science and Engineering

Academic Year 2023 – 2024 (Odd Semester)

Degree, Semester & Branch : V Semester B.E CSE
Course Code & Title : CCS334 & Big Data Analytics
Name of the Faculty member : Dr.I.Gethzi Ahila Poornima, AP(SG) / CSE

Innovative Practice Description

- **Unit / Topic:** Unit III /Map Reduce
- **Course Outcome:** CO3
- **Activity Chosen:** Flipped Classroom
- **Justification:**

Inverting the classroom is a teaching strategy known as "flipping the classroom." This method "flipped" typical in-class instruction to better serve the needs of each individual student. Map Reduce Concept is one of the best parallel processing concept. By dividing the data into smaller portions that will be handled in parallel by numerous commodity computers, it enables concurrent processing and reduces processing time. Since parallel processing is one of the necessary things, the learner needs to understand the internal processes that Map Reduce employs.

- **Time Allotted for the Activity:** 40 Minutes

- **Details of the Implementation:**

A flipped classroom is a kind of cooperative learning in which students watch videos and refer notes uploaded in LMS at home, take notes based on their comprehension level, and then solve the given problem using the knowledge they have learned. As a result, students' self-learning habits get better. The happening of event was intimated to the students in the class room exactly one week in prior. The supporting notes and video (<https://www.youtube.com/watch?v=mNP44rZYiAU>) were uploaded five days before in LMS. On the day of event, totally 25 students were present. Also, teams of size five were formed to discuss the concept and discussed in the class room. The topic and the event was announced to them in the previous week itself. After discussion with the team members, one per team was called on stage to present their ideas related to Map reduce. Figure 1 shows a sample snap taken during the event and it covers three teams.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO10	PSO1
CO3	2	3	3	3

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO10	PSO1
	2	2	2	2
Justification for correlation	The students will be able to know the distribution of works among various nodes. problem	The students will be able to analyze various complex Engineering problems especially business domains by understanding Mao Reduce concept.	The students will work effectively and demonstrate effective communication, problem-solving skills	The student will analyze huge amount of data to solve complex computation task in efficient way.

- **Images / Screenshot of the practice:**



Figure 1. Three Teams represented

- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

- They enjoyed the freedom to learn at their own pace
- They thought that independent learning would enable them to experiment with new concepts.

- ❖ ***Benefit of the practice:***

- Students learn at their own pace;
- There is more time for one-on-one interaction between the teacher and the student;
- There is more opportunity for student participation.
- It encourages students to arrive prepared to class.

- ❖ ***Challenges faced in implementation:***

- If the students do not prepare for the event, then the time is entirely waster.
- The depth of the subject can be dictated by the student themselves or the group the student is working.
- The time and effort required from a teacher's perspective initially when creating the flipped class material is higher than for a traditional class.

References:

- ❖ <https://www.teachthought.com/learning/the-definition-of-the-flipped-classroom/#:~:text=A%20flipped%20classroom%20is%20a,the%20students%20independently%20at%20home>.
- ❖ https://en.wikipedia.org/wiki/Flipped_classroom
- ❖ [youtube.com/watch?v=BCIxikOq73Q](https://www.youtube.com/watch?v=BCIxikOq73Q)
- ❖ <https://omerad.msu.edu/teaching/teaching-skills-strategies/27-teaching/162-what-why-and-how-to-implement-a-flipped-classroom-model>
- ❖ https://www.ritrjpm.ac.in/images/computer-science/2022-2023/3_GS_CS8592_Flipped_classroom.pdf
- ❖ https://www.ritrjpm.ac.in/images/computer-science/2022-2023/5.Flipped_Classroom_FDS.pdf

Signature of Faculty Member

HOD